

CLEF: Cluster-Layer Entropy Focus

v1.0 - Public Theoretical Framework

Boundary Ledgers, Checked Coherence, Recursive Coupling, and Worked Toy Accounting

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Status. Public theoretical framework, version 1.0. CLEF is presented as a layer-explicit boundary-measurement language with default multi-channel weighting rules, proxy-uncertainty rules, recursive boundary-coupling rules, a scale-relevance stopping rule, and a worked numerical toy example. Experimental routes remain future refinement paths unless data are reported.

Distinction seals; fold-thinning leaks; pop ruptures.

Surface area counts the boundary. Curvature tells where the boundary is stressed.

Contact tells what the boundary grabs.

The boundary ledger is not a free-energy primitive. It is declared accounting for boundary costs.

A measurement layer makes a distinction accountable.

Abstract

Cluster-Layer Entropy Focus (CLEF) provides a formal boundary-measurement language for describing how distinctions become readable at declared layers. This theoretical paper establishes layer declarations, coherence-normalization rules, boundary-state ladders, and a cost family in which excess cost functions as the keystone variable connecting readable boundaries to later relaxation, leak, pop, damping, resonance, or trace export.

CLEF separates a metaphysical distinction, understood as a named difference, from a scientific distinction, understood as a named difference with a declared measurement layer. The minimum public layer declaration is

$$L_{\min} = (X, r, N_L, \mathcal{A}_L, W_L), \quad (1)$$

where X is the observed field, r is aperture or scale, N_L is the noise floor, $\mathcal{A}_L$ is the extraction method, and W_L is the readability wall. A layer fails when the declared extraction method cannot produce a readable boundary above the noise floor at the declared wall. Dynamic, perturbation-based, resonance, relaxation, cadence, or pop claims require additional declarations such as a temporal window or sampled-time set τ/T_L , perturbation Π_L , and medium/apparatus path $\mathcal{M}_L$. A dynamic CLEF test may operate on discrete snapshots rather than continuous real-time fluid simulation.

The central CLEF score compares normalized coherence gained at layer L to the cost required to create, reveal, hold, or read the corresponding distinction:

$$\text{CLEF}_L = \frac{\Delta\Phi_L}{\mathcal{K}_L}. \quad (2)$$

The main layer- L score remains a creation/readability score,

$$\text{CLEF}_{q,L}^{\text{create}} = \frac{\Delta\Phi_{q,L}}{\mathcal{K}_{q,L}^{\text{create}}}, \quad (3)$$

while the excess term is mandatory for later boundary behavior:

$$\mathcal{K}_L^\Delta = [\mathcal{K}_L^\sigma - \mathcal{K}_L^{\min}]_+. \quad (4)$$

$\mathcal{K}_L^\Delta$ measures stored boundary cost above the accessible closure baseline and explains why a created boundary can later relax, leak, fold-thin, pop, damp, resonate, or export a trace.

CLEF is a theoretical framework proposal rather than a completed empirical validation. It is layered over established physical and mathematical work instead of replacing it: surface tension, curvature pressure, capillary length, contact angles, thermodynamic potentials, thin-film rupture, curvature elasticity, image segmentation, topology, tribology, sloshing, acoustic sensing, and cellular biology remain governed by their existing scholarly literatures. CLEF does not inherit free energy as a primitive; it inherits the discipline of accounting for boundary costs. The operative object is a boundary ledger: a declared set of terms that make a boundary readable, costly, stable, unstable, excess-loaded, or available for closure. Curvature is diagnostic by default: in many physical ledgers it already appears through surface tension, pressure balance, wetting, wall constraints, or neighboring-boundary conditions. It enters the ledger as a separate term only when a neighboring-ledger residual, bending elasticity, wall pinning, roughness, or unresolved curvature-stress role is explicitly declared.

Multi-channel reports show the normalized coherence vector before any scalar reduction. When no domain-specific weighting is justified, scalar reports use equal weights across independent evidence families rather than equal weights across every raw channel; required validity checks act as gates, not averageable evidence. Boundary interactions are represented by declared couplings, residuals, bottlenecks, or gates, not hidden in arbitrary weights. Proposed oil/water cells, lava-lamp proxy video, sealed-container tap response, and classroom emulsification demonstrations are retained as possible experimental routes for future refinement.

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1 Layer declaration

A CLEF layer is the declared system that makes a boundary readable. v1.0 separates a metaphysical distinction from a scientific distinction:

$$\text{metaphysical distinction} = \text{named difference}, \quad (5)$$

$$\text{scientific distinction} = \text{named difference with declared measurement layer}. \quad (6)$$

The minimum layer declaration for a public CLEF test is:

$$\boxed{L_{\min} = (X, r, N_L, \mathcal{A}_L, W_L)}. \quad (7)$$

The terms are:

- X : observed field, image, signal, state, material boundary, or symbolic stream;
- r : aperture, blur radius, sampling scale, or resolution;
- N_L : noise floor, threshold model, or minimum readable contrast; N_L may be scalar or vector-valued when the declared noise model has multiple channels;
- $\mathcal{A}_L$: extraction, segmentation, measurement, or reading method;
- W_L : readability wall where the source becomes trace-readable or fails to return as trace.

The operational failure condition is:

A layer fails when the declared extraction method cannot produce a readable boundary above the noise floor at the declared readability wall.

This minimum can be read as a measurement geometry rather than geometry in the narrow shape-only sense:

$$G_L = (X, r, W_L), \quad (8)$$

with N_L and $\mathcal{A}_L$ declaring what counts above noise and how the boundary is extracted. In this sense, geometry means the declared relation among field, aperture, wall, and readable trace.

For dynamic, regime-sensitive, or perturbation-based claims, the full layer expands to:

$$L_{full} = (X, r, \tau, N_L, \mathcal{A}_L, R_L, W_L, \Pi_L, \mathcal{M}_L), \quad (9)$$

where:

- τ is the temporal window or duration;
- R_L is the physical, formal, or semantic regime being assumed;
- Π_L is the probe, tap, perturbation, or intervention;
- $\mathcal{M}_L$ is the medium/apparatus path returning the trace.

For a public dynamic CLEF test, τ , Π_L , and $\mathcal{M}_L$ are required by default. If the claim includes time, perturbation, resonance, relaxation, cadence, slosh, reaction, or pop behavior, time and return path cannot be omitted. However, this does not require continuous real-time simulation. A dynamic CLEF test may be sampled-time:

$$T_L = \{t_0, t_1, t_2, \dots, t_n\}, \quad \tau_L = [t_0, t_n], \quad (10)$$

with layer-readable states

$$S_L(t_i) = \mathcal{A}_L(X(t_i); r, N_L, W_L), \quad (11)$$

and reported transitions

$$S_L(t_i) \rightarrow S_L(t_{i+1}). \quad (12)$$

If no deliberate perturbation is applied, Π_L should be declared as ambient, null, or uncontrolled rather than omitted. The guiding rule is:

If the claim is dynamic, time cannot disappear; the unresolved flow between sampled times may remain unresolved.

This gives a compact reading operator:

$$\mathcal{R}_L : S \longrightarrow Y_L, \quad (13)$$

where S is the source system and Y_L is the layer-readable output. Two layers may read the same source differently:

$$\mathcal{R}_{L_1}(S) \neq \mathcal{R}_{L_2}(S). \quad (14)$$

This is not a measurement error by itself. It is the starting condition of CLEF.

1.1 Trace, wall, and unitization

The source-depth is not assigned units simply because it exists. Units enter through a declared boundary effect:

$$T_L(C) = \text{trace}_L(C \rightarrow W_L), \quad (15)$$

where C is source-depth and W_L is the measurement wall.

The unitization rule is:

$$\text{source concept: unitless before wall,} \quad \text{boundary effect: unitized after wall.} \quad (16)$$

Thus:

$$\text{concept} \neq \text{measured boundary effect.} \quad (17)$$

Unitless does not mean meaningless; it means the concept has not yet passed through a declared measurement wall. A concept may be meaningful before measurement; CLEF becomes accountable only when the effect is declared and can fail to match the model.

2 Boundary-state ladder

The minimal boundary-state spine is:

$$D_L \rightarrow F_L \rightarrow P_L. \quad (18)$$

It belongs near the front because it tells the reader what later costs and traces are being used to describe.

State	Meaning	Layer condition	Trace behavior
D_L	sealed distinction / readable boundary	$C_{T,L} \leq \theta_{T,L}$	no exported trace
F_L	fold-thinning leak / leaky, unbroken boundary	$C_{T,L} > \theta_{T,L}$ and $\Delta B_L < \theta_{\text{rupture},L}$	weak, local, or non-propagating trace
P_L	pop / rupture-release	$\Delta B_L \geq \theta_{\text{rupture},L}$	strong propagated trace

Here $C_{T,L}$ is a scalar trace-coupling score at layer L unless a vector-valued trace channel is explicitly declared. ΔB_L is the boundary-update magnitude and $\theta_{\text{rupture},L}$ is the layer-relative rupture threshold.

The short ladder gives the minimal closure states. A longer interaction ladder distinguishes absorption and rotation before leak and rupture:

$$D_L \rightarrow A_L \rightarrow R_L \rightarrow F_L \rightarrow P_L. \quad (19)$$

Here A_L is one-way absorption/flux and R_L is loop, rotation, circulation, or return structure. In prose:

Distinction cuts. Absorption lands. Rotation returns. Fold-thinning leaks. Pop releases.

3 Core CLEF score

The general CLEF score is:

$$\text{CLEF}_L = \frac{\Delta\Phi_L}{\mathcal{K}_L}. \quad (20)$$

Here $\Delta\Phi_L$ is the layer-normalized coherence gained, and $\mathcal{K}_L$ is the declared cost associated with creating, revealing, holding, or reading the boundary distinction.

Because coherence can be measured through multiple channels, introduce a channel index q :

$$\text{CLEF}_{q,L} = \frac{\Delta\Phi_{q,L}}{\mathcal{K}_{q,L}}. \quad (21)$$

Example channels include segmentation, mutual information, prediction, phase coherence, recurrence, and external response. These should not be collapsed before normalization.

A vector form is:

$$\vec{\Phi}_L = [\Phi_{\text{seg},L}, \tilde{\Phi}_{MI,L}, \Phi_{\text{pred},L}, \Phi_{\text{phase},L}, \tilde{\Phi}_{\text{auto},L}], \quad (22)$$

with scalar aggregate:

$$\Delta\Phi_L^* = \sum_i w_i \Delta\tilde{\Phi}_{i,L}, \quad \sum_i w_i = 1, \quad w_i \geq 0. \quad (23)$$

The aggregate score is then:

$$\text{CLEF}_L^* = \frac{\Delta\Phi_L^*}{\mathcal{K}_L}. \quad (24)$$

The weights are not universal. They are part of the declared layer. If no scalar reduction is justified, the vector should remain the report. When scalar reduction is required, CLEF uses the default grouping rule defined in Section 4.2: independent evidence families may be weighted equally; dependent channels must be grouped before averaging; gates cannot be averaged away.

4 Coherence channels

4.1 Default normalization protocol

CLEF does not require every coherence channel to share the same raw units. Instead, every channel j must declare a raw observable $Q_{j,L}$, a null or baseline value $Q_{j,L}^{\text{null}}$, a reference or saturation value $Q_{j,L}^{\text{ref}}$, and a normalized score:

$$\tilde{\Phi}_{j,L} = \text{clip}_{[0,1]} \left(\frac{Q_{j,L} - Q_{j,L}^{\text{null}}}{Q_{j,L}^{\text{ref}} - Q_{j,L}^{\text{null}}} \right). \quad (25)$$

The reported channel change is:

$$\Delta\tilde{\Phi}_{j,L} = \tilde{\Phi}_{j,L}(t_1) - \tilde{\Phi}_{j,L}(t_0), \quad (26)$$

and the aggregate gain is:

$$\Delta\Phi_L = \sum_j w_j \Delta\tilde{\Phi}_{j,L}, \quad \sum_j w_j = 1, \quad w_j \geq 0. \quad (27)$$

Segmentation, raw-field support, cadence, resonance, prediction, phase coherence, ledger-relaxation consistency, and external response may all serve as channels, provided each is normalized before combination.

Normalization lets segmentation, cadence, resonance, and prediction speak the same CLEF language without pretending they are the same measurement.

4.2 Default weighting, gates, and dependent channels

CLEF reports the coherence vector before scalar reduction:

$$\vec{\Phi}_L = [\tilde{\Phi}_{1,L}, \tilde{\Phi}_{2,L}, \dots, \tilde{\Phi}_{m,L}]. \quad (28)$$

A scalar score is optional. When no domain-specific weighting is justified, equal weighting is used across independent evidence families, not blindly across raw channels. If G_g is an evidence family and G is the number of independent families, the neutral scalar is:

$$\Phi_L^{\text{neutral}} = \frac{1}{G} \sum_{g=1}^G \left(\frac{1}{|G_g|} \sum_{j \in G_g} \tilde{\Phi}_{j,L} \right). \quad (29)$$

This prevents related channels from double-counting the same boundary read. For example, segmentation IoU, segmentation shear, and raw-field edge support belong to the same visual-boundary family and must be grouped rather than weighted independently; resonance may belong to a separate response family.

If a channel is required for the claim to be valid, it is a gate rather than an averageable weight:

$$\mathcal{G}_L = \prod_a H(\tilde{\Phi}_{a,L} - \theta_{a,L}). \quad (30)$$

A failed gate does not become a small negative preference; it means the reported claim is not supported at the declared layer. A gated scalar may be written:

$$\Phi_L^{\text{checked}} = \mathcal{G}_L \Phi_L^{\text{neutral}}. \quad (31)$$

Interacting boundary limits are not explained by weights. If two limits physically shape the same field, their interaction belongs in a declared coupling, residual, bottleneck, or gate:

$$\mathcal{L}_L = \mathcal{L}_{a,L} + \mathcal{L}_{b,L} + \mathcal{R}_{ab,L}. \quad (32)$$

The reporting rule is:

Weights summarize evidence; they do not explain interaction. Do not let a good channel average away a failed boundary.

4.3 Segmentation coherence

Let S_L be a segmentation at layer L . If two readings are used, for example hierarchical contour segmentation S_L^H and density-stability clustering S_L^C , define intersection-over-union for pieces A, B :

$$\mathcal{J}(A, B) = \frac{|A \cap B|}{|A \cup B|}. \quad (33)$$

For complete segmentations, a simple agreement score is:

$$\mathcal{J}_L(S_L^H, S_L^C) = \frac{1}{K} \sum_{k=1}^K \max_j \frac{|P_k^H \cap P_j^C|}{|P_k^H \cup P_j^C|}. \quad (34)$$

The segmentation shear is:

$$\Delta_{\text{shear},L} = 1 - \mathcal{J}_L(S_L^H, S_L^C). \quad (35)$$

A pass gate can be declared:

$$\Delta_{\text{shear},L} \leq \varepsilon_{\text{shear},L}. \quad (36)$$

Layer stability under small aperture changes is computed on segmentation masks or declared boundary pieces, not directly on raw images:

$$\text{Stab}(S_L) = \frac{1}{|\mathcal{N}(r)|} \sum_{\delta r \in \mathcal{N}(r)} \mathcal{J}(S_r, S_{r+\delta r}). \quad (37)$$

A segmentation coherence channel can be:

$$\Phi_{\text{seg},L} = \text{Stab}(S_L^H) \text{Stab}(S_L^C) \mathcal{J}_L(S_L^H, S_L^C). \quad (38)$$

A single segmentation channel can demonstrate that a boundary can be extracted. It is not, by itself, a checked CLEF report. v1.0 uses the rule:

$$\vec{\Phi}_L^{\text{toy}} = [\Phi_{\text{seg},L}], \quad (39)$$

for an illustrative single-channel toy, but

$$\vec{\Phi}_L^{\text{checked}} = [\Phi_{\text{seg},L}, \Phi_{\text{check},L}] \quad (40)$$

for a checked public report. The second channel must not merely repackage segmentation:

$$\Phi_{\text{check},L} \not\equiv f(\Phi_{\text{seg},L}). \quad (41)$$

A simple visual check is raw-field boundary support:

$$\Phi_{\text{edge},L}(t) = \frac{1}{|\partial S_L(t)|} \int_{\partial S_L(t)} |\widetilde{\nabla X}| ds, \quad (42)$$

which asks whether the extracted boundary sits on contrast or gradient support in the raw field. In prose:

One channel can find a boundary. Two channels can begin to check it.

4.4 Mutual information and interaction

Mutual information is a coupling measure, not by itself a proof of interaction:

$$I(X; Y) = \sum_{x,y} p(x,y) \log \frac{p(x,y)}{p(x)p(y)}. \quad (43)$$

Normalize by available entropy:

$$\tilde{\Phi}_{MI,L} = \frac{I(X; Y)}{\min\{H(X), H(Y)\}}, \quad (44)$$

when the denominator is nonzero.

CLEF distinguishes coupling from perturbation-sensitive trace. A minimal interaction criterion is:

$$d(Y(X_1), Y(X_0)) > \theta_{I,L}. \quad (45)$$

Thus:

$$\text{mutual information} \neq \text{interaction by itself}. \quad (46)$$

4.5 Prediction, phase, and recurrence

Predictive gain can be normalized as:

$$\Phi_{\text{pred},L} = \left[1 - \frac{\mathcal{E}_{\text{model},L}}{\mathcal{E}_{\text{baseline},L}} \right]_+ . \quad (47)$$

For an oscillatory or cadence-bearing signal with phases $\theta_j(t)$, phase coherence may be represented by:

$$\Phi_{\text{phase},L} = \left| \frac{1}{n} \sum_{j=1}^n e^{i\theta_j(t)} \right|. \quad (48)$$

Autocorrelation can be rescaled from $[-1, 1]$ to $[0, 1]$:

$$\tilde{\Phi}_{\text{auto},L} = \frac{1 + \rho_L(\Delta t)}{2}. \quad (49)$$

A return-period rule for a normalized state z_n is:

$$p^* = \min\{p > 0 : d(z_{n+p}, z_n) \leq \varepsilon_L\}, \quad (50)$$

with irreducibility condition:

$$d(z_{n+q}, z_n) > \varepsilon_L, \quad 0 < q < p^*. \quad (51)$$

4.6 Cadence and resonance channels

Cadence claims require a declared discrete sampling set or sampling rule. Cadence is defined on discrete sampled states; it is not treated here as a continuous-time invariant. Cadence is treated as a return-pattern coherence channel. Given sampled states $S_L(t_i)$ and a candidate discrete period p , define an average return distance:

$$d_p = \frac{1}{n-p} \sum_{i=0}^{n-p} d(S_L(t_i), S_L(t_{i+p})). \quad (52)$$

Normalize cadence coherence by a layer-declared return tolerance $\varepsilon_{\text{cad},L}$:

$$\Phi_{\text{cad},L}(p) = 1 - \text{clip}_{[0,1]} \left(\frac{d_p}{\varepsilon_{\text{cad},L}} \right), \quad (53)$$

with reported cadence channel

$$\Phi_{\text{cad},L} = \max_p \Phi_{\text{cad},L}(p). \quad (54)$$

An irreducible cadence requires that shorter returns fail the same tolerance:

$$d_q > \varepsilon_{\text{cad},L}, \quad 0 < q < p^*. \quad (55)$$

Resonance is treated as response-concentration coherence after a declared perturbation. Given returned signal $y_L(t)$ and spectrum

$$Y_L(f) = \mathcal{F}[y_L(t)], \quad (56)$$

define the fraction of power concentrated in a declared band around f^* :

$$\Phi_{\text{res},L} = \frac{\int_{f^*-\delta}^{f^*+\delta} |Y_L(f)|^2 df}{\int_{\mathcal{F}_L} |Y_L(f)|^2 df}. \quad (57)$$

This channel is already bounded in $[0, 1]$ when the denominator is nonzero. A resonance claim requires τ or T_L , Π_L , and $\mathcal{M}_L$ because the returned signal is a time-dependent response through a declared medium/apparatus path.

4.7 Coherence-channel summary table

The following table is a compact map of the default CLEF coherence channels. Each channel must declare its raw observable, normalization path, and role before entering $\Delta\Phi_L$.

Channel	Raw observable	Normalization	Role
Segmentation	IoU, shear, mask stability	$[0, 1]$ by IoU/stability	boundary readability
Mutual information	$I(X; Y)$ in bits or nats	divide by available entropy	cross-layer relation/coupling
Prediction	model error vs. baseline error	$1 - \mathcal{E}_{model}/\mathcal{E}_{base}$ clipped	dynamic utility
Phase	circular mean of phases	already in $[0, 1]$	synchrony / cadence support
Autocorrelation	$\rho_L(\Delta t)$	$(1 + \rho)/2$	memory / return tendency
Cadence	sampled return distance d_p	$1 - d_p/\varepsilon_{cad,L}$ clipped	recurrence
Resonance	spectral power concentration	band power / total power	response after perturbation
Raw-field support	gradient, edge, or contrast along ∂S_L	declared base-line/reference	extraction check

A single-channel toy may report only one row. A checked public CLEF report should report at least two non-equivalent channels.

5 Cost family: creation, state, excess, and keep

v1.0 resolves the cost-symbol family. The base symbol $\mathcal{K}$ is retained because it cleanly signals cost. Superscripts specify the role.

5.1 Creation cost

Creation cost is the cost required to make a distinction readable:

$$\mathcal{K}_L^{\text{create}}(s_0 \rightarrow s_1). \quad (58)$$

For an interface-area transition:

$$\mathcal{K}_L^{\text{create}}(s_0 \rightarrow s_1) = [\mathcal{L}_L(s_1) - \mathcal{L}_L(s_0)]_{+, \text{creation}}, \quad (59)$$

or, for a shallow two-phase interface with interfacial tension γ_{12} ,

$$\mathcal{K}_{12,L}^{\text{create}} = \gamma_{12} \Delta A_{12,L}. \quad (60)$$

The creation/readability score is:

$$\text{CLEF}_{q,L}^{\text{create}} = \frac{\Delta\Phi_{q,L}}{\mathcal{K}_{q,L}^{\text{create}}}. \quad (61)$$

This remains the main score because it answers: how much layer-readable coherence was gained per creation cost?

5.2 State cost

State cost is the current boundary-ledger value:

$$\mathcal{K}_L^\sigma(s) = \mathcal{L}_L(s). \quad (62)$$

The superscript σ is used because state, store, configuration, and system condition are familiar computational readings of σ .

5.3 Closure baseline

The accessible closure baseline is not always a global thermodynamic minimum. It is a layer-relative constrained baseline:

$$\mathcal{K}_L^{\min} = \inf_{s \in \Omega_L^{\text{adm}}} \mathcal{L}_L(s), \quad (63)$$

where Ω_L^{adm} is the admissible state set under the layer's constraints. Constraints may include volume, wall geometry, wetting, temperature, boundary conditions, material compatibility, available time, and the chosen reading layer.

5.4 Excess cost as keystone

The v1.0 keystone is:

$$\boxed{\mathcal{K}_L^\Delta = [\mathcal{K}_L^\sigma - \mathcal{K}_L^{\min}]_+}. \quad (64)$$

$\mathcal{K}_L^\Delta$ is not optional. It is the one-layer-up state variable that explains why a created boundary has stored consequence. Creation gives a readable boundary. Excess says whether that boundary is loaded.

In words:

$$\mathcal{K}^{\text{create}} = \text{cost to make the readable boundary}, \quad (65)$$

$$\mathcal{K}^\Delta = \text{cost still stored above accessible closure baseline}. \quad (66)$$

The chain is:

$$\mathcal{K}_L^\Delta > 0 \quad \Rightarrow \quad \text{available relaxation, leak, fold-thinning, pop, damping, resonance shift, or trace export}. \quad (67)$$

This implication is not deterministic. It says that excess is the state variable to inspect before expecting later boundary behavior.

5.5 Proxy and physical units for excess cost

$\mathcal{K}_L^\Delta$ may be reported in proxy units for theoretical, toy, image-based, classroom, or first-pass CLEF tests. Physical units are required when the claim is scale-bearing: when the value is meant to compare across setups, predict material behavior, estimate real energy/work/power, or support engineering-scale interpretation.

Proxy-unit reports must be labeled as proxy units in the text, tables, and captions. A proxy excess report has the form:

$$\mathcal{K}_L^{\Delta, \text{proxy}} = [Q_L - Q_L^{\min}]_+, \quad (68)$$

where Q_L may be pixel interface length, normalized roughness, segmentation boundary length, droplet count, resonance-band concentration, or another declared proxy.

A calibrated quasi-two-dimensional two-phase cell may instead report physical excess:

$$\mathcal{K}_{12,L}^{\Delta, \text{phys}} = \gamma_{12} h [\ell_L - \ell_{\min, L}]_+, \quad (69)$$

where γ_{12} is interfacial tension, h is cell depth, and ℓ_L is physical interface length. Thus:

$$\text{proxy report} \Rightarrow \text{pattern-readable within declared layer}, \quad (70)$$

$$\text{physical report} \Rightarrow \text{scale-bearing across declared material system}. \quad (71)$$

Proxy units can show that a boundary is loaded; physical units are needed when the loading must be carried across scale.

5.6 Proxy uncertainty rule

Proxy reports do not need full statistical machinery merely because a value is pixel-based or manually counted. They need enough uncertainty notation to decide whether the claim survives the declared layer. A simple proxy report may be written:

$$Q_L^{\text{proxy}} = q \pm \delta q, \quad (72)$$

where the unit is explicitly labeled as proxy units, pixels, counts, masks, normalized scores, or another declared proxy. For a difference between two sampled states:

$$\Delta Q_L^{\text{proxy}} = Q_L^{\text{proxy}}(t_1) - Q_L^{\text{proxy}}(t_0), \quad (73)$$

a conservative uncertainty is:

$$\delta(\Delta Q_L) \approx \delta Q_L(t_1) + \delta Q_L(t_0). \quad (74)$$

A proxy change is supported only if it remains above both uncertainty and the noise floor:

$$|\Delta Q_L^{\text{proxy}}| > \delta(\Delta Q_L) + N_L. \quad (75)$$

If instead:

$$|\Delta Q_L^{\text{proxy}}| \leq \delta(\Delta Q_L) + N_L, \quad (76)$$

the result is ambiguous at the declared layer. In prose:

Uncertainty stops being background noise when it can change the layer-readable answer.

5.7 Keep cost

Maintenance is better treated as rate or accumulated work. Define keeping power:

$$\mathcal{P}_L^{\text{keep}}(t) = \text{power required to hold excess structure away from relaxation}. \quad (77)$$

Then accumulated keep cost is:

$$\mathcal{K}_{L,[t_0,t_1]}^{\text{keep}} = \int_{t_0}^{t_1} \mathcal{P}_L^{\text{keep}}(t) dt. \quad (78)$$

If no active holding is applied, $\mathcal{K}^{\text{keep}}$ may be zero even when $\mathcal{K}^\Delta > 0$.

5.8 Consolidated cost-family table

The cost family is easiest to read as a hierarchy. Creation cost reports how readability is produced. State and baseline costs locate the current boundary in its ledger. Excess cost is the mandatory keystone for later behavior. Keep cost records work required to hold the excess away from relaxation.

Symbol	Name	Role
$\mathcal{K}_L^{\text{create}}$	creation cost	Cost required to make a boundary readable at layer L .
$\mathcal{K}_L^\sigma$	state cost	Current boundary-ledger value: $\mathcal{K}_L^\sigma(s) = \mathcal{L}_L(s)$.
$\mathcal{K}_L^{\min}$	constrained baseline	Lowest accessible ledger value under declared constraints: $\mathcal{K}_L^{\min} = \inf_{s \in \Omega_L^{\text{adm}}} \mathcal{L}_L(s)$.
$\mathcal{K}_L^\Delta$	excess / unclosed cost	Mandatory keystone: $\mathcal{K}_L^\Delta = [\mathcal{K}_L^\sigma - \mathcal{K}_L^{\min}]_+$. This reports stored boundary cost available to drive closure, fold-thinning, leak, pop, damping, resonance shift, or exterior trace.
$\mathcal{P}_L^{\text{keep}}(t)$	keeping power	Instantaneous power required to hold excess structure away from relaxation.
$\mathcal{K}_{L,[t_0,t_1]}^{\text{keep}}$	maintenance / keeping cost	Accumulated keep work: $\mathcal{K}_{L,[t_0,t_1]}^{\text{keep}} = \int_{t_0}^{t_1} \mathcal{P}_L^{\text{keep}}(t) dt$.

The compact relationship is:

$\mathcal{K}_L^{\text{create}}$ makes the boundary readable; $\mathcal{K}_L^\Delta$ says how much of that boundary remains loaded after readability appears.

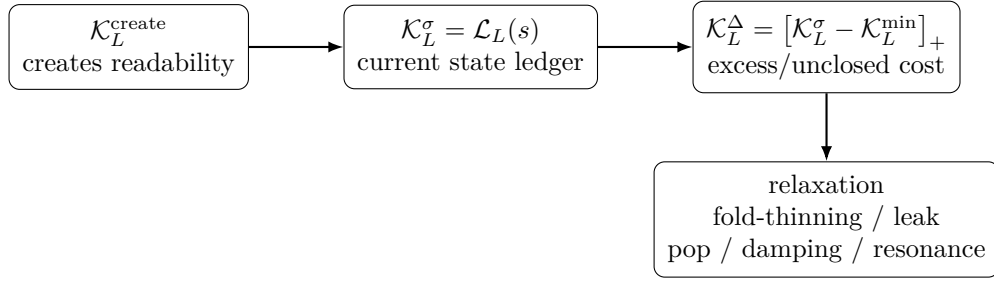


Figure 1: CLEF state-cost ladder. Creation cost reports how a boundary becomes readable. State cost locates the current boundary in its ledger. Excess cost is the one-layer-up keystone that can later appear as relaxation, fold-thinning, leak, pop, damping, resonance shift, or exterior trace.

6 Boundary ledger, not free-energy primitive

CLEF does not inherit free energy as a primitive; it inherits the discipline of accounting for boundary costs. A boundary ledger is a declared set of terms that make a boundary readable, costly, stable, unstable, excess-loaded, or available for closure.

Friction/contact remains a footnote-level bridge rather than a full CLEF branch.¹

This phrasing is deliberate. In thermodynamics and related fields, free-energy formalisms provide powerful tools for tracking interfacial energy, gravitational potential, wetting/contact contributions, curvature-elastic terms, chemical potential, and relaxation tendencies. CLEF preserves the usable accounting discipline of those approaches but names its operative object more narrowly: a boundary ledger. Where CLEF overlaps established thermodynamic potentials, the overlap should be cited directly; CLEF does not treat “free energy” as a universal explanatory object.

A boundary ledger is layer-declared. Its general form is:

$$\mathcal{L}_L(s; \Theta_L) = \sum_{i=1}^m \alpha_{i,L} T_{i,L}(s), \quad (79)$$

¹Friction/contact is used only as a boundary-interaction analogy. CLEF does not reduce tribology to surface tension; real friction remains governed by contact mechanics.

where $T_{i,L}$ are declared terms and $\alpha_{i,L}$ are physical constants, modeling weights, or layer-normalization coefficients. The ledger is valid only after its terms and interpretation are declared.

A term is not a ledger term merely because it is measurable. It enters the boundary ledger only when its physical contribution or declared proxy role is stated.

Status	CLEF role	Symbols / examples	Ledger?	Rule
Physical ledger term	Established physical contribution	γA , $\rho g z$, contact/wetting terms, calibrated bending energy	yes	Use when material constants, geometry, and units are declared.
Modeling proxy term	Declared approximation	pixel length, normalized roughness, λC_L , $\mathcal{K}_L^{\Delta, \text{proxy}}$	sometimes	State what physical or unresolved effect it approximates.
Diagnostic term	Readable signal, not cost by default	C_L , segmentation shear, cadence return, resonance shift, edge support	no, unless promoted	May guide interpretation; promotion requires explicit proxy/physical role.
Layer-declaration term	Defines how the boundary is read	$X, r, N_L, \mathcal{A}_L, W_L, \tau, \Pi_L, \mathcal{M}_L$	no	Required for measurement accountability and reproducibility; not itself a boundary cost.

6.1 Boundary conditions, coupling, and recursive promotion

Boundary conditions may coexist without directly acting on each other, but they become coupled when the same field must satisfy them. A generic field problem has the form:

$$\mathcal{D}u = f \quad \text{inside } \Omega, \quad (80)$$

with boundary rules:

$$\mathcal{B}_1[u] = g_1 \quad \text{on } \Gamma_1, \quad \mathcal{B}_2[u] = g_2 \quad \text{on } \Gamma_2. \quad (81)$$

The boundary conditions do not need to interact as objects. They are coupled through the solution u that must satisfy them together.

In CLEF ledger form, two boundary ledgers plus a coupling or residual can be written:

$$\mathcal{L}_L = \mathcal{L}_{a,L} + \mathcal{L}_{b,L} + \mathcal{R}_{ab,L}. \quad (82)$$

If the residual becomes readable above a declared threshold,

$$|\mathcal{R}_{ab,L}| > \theta_{R,L}, \quad (83)$$

then the interaction may be promoted to the next layer as an observed field:

$$\mathcal{R}_{ab,L} \rightarrow X_{L+1}. \quad (84)$$

The recursion stops operationally when the residual falls below the noise floor, is already explained by declared terms, reaches the readability wall, or adds no new coherence:

$$|\mathcal{R}_{ab,L}| \leq N_L, \quad \text{or} \quad \mathcal{R}_{ab,L} \subset \mathcal{L}_L^{\text{known}}, \quad \text{or} \quad W_L, \quad \text{or} \quad \Delta\Phi_{L+1} \approx 0. \quad (85)$$

Thus:

A boundary interaction becomes a new field only when the interaction itself becomes readable. The recursion ends where readability stops, or where a higher layer adds no new coherence.

A diagnostic term becomes a ledger term only when its contribution is required to explain a mismatch between declared ledger terms and observed boundary behavior.

For a two-phase liquid-wall system, a physical ledger may include:

$$\mathcal{L}_{12,L} = \gamma_{12}A_{12,L} + \gamma_{1s}A_{1s,L} + \gamma_{2s}A_{2s,L} + \int_{\Omega_1} \rho_1 g z dV + \int_{\Omega_2} \rho_2 g z dV + \mathcal{C}_{wall,L}. \quad (86)$$

The terms are:

- $\gamma_{12}A_{12,L}$: phase-1/phase-2 interface contribution;
- $\gamma_{1s}A_{1s,L}$ and $\gamma_{2s}A_{2s,L}$: wall contact / wetting terms;
- $\int \rho g z dV$: gravity-density contribution;
- $\mathcal{C}_{wall,L}$: container geometry, roughness, pinning, constraint, and wall-grabbiness terms.

Where a physical example overlaps established thermodynamic potentials, that overlap should be cited directly. CLEF's claim is narrower:

$$\text{readable boundaries carry ledgers.} \quad (87)$$

Some ledger terms are established energy terms. Others are measurement-layer proxies. Treating all of them as “free energy” would erase that distinction.

7 Geometry of boundary cost

The minimal geometry is not only volume or surface area. CLEF uses three progressively stronger descriptors:

$$V_L = \text{Vol}(\Omega_L) \quad (\text{enclosed amount}), \quad (88)$$

$$A_L = \text{Area}(\partial\Omega_L) \quad (\text{amount of boundary}), \quad (89)$$

$$C_L = \text{curvature distribution diagnostic} \quad (\text{how boundary bends}). \quad (90)$$

A compactness ratio such as:

$$\frac{V_L}{A_L} \quad (91)$$

can compare gross shape efficiency, but it does not determine shape. Two shapes can share volume and surface area while differing in folds, necks, curvature concentration, topology, contact line, and rupture risk.

The missing third term is curvature distribution. For a smooth surface with principal curvatures κ_1, κ_2 , mean curvature is:

$$H = \frac{\kappa_1 + \kappa_2}{2}. \quad (92)$$

A diagnostic curvature proxy is:

$$C_L = \int_{\partial\Omega_L} H^2 dA. \quad (93)$$

This proxy should not automatically become a separate physical ledger term. Curvature is diagnostic by default because, in many physical ledgers, curvature already appears through surface tension, pressure balance, wetting, wall constraints, or neighboring-boundary conditions. Adding it again as an independent ledger cost would double-count unless a residual/proxy role is declared. For ordinary fluid-fluid interfaces, curvature appears through surface tension and pressure. The Young-Laplace relation is:

$$\Delta p = \gamma \left(\frac{1}{R_1} + \frac{1}{R_2} \right) = \gamma(\kappa_1 + \kappa_2) = 2\gamma H. \quad (94)$$

Thus curvature often marks where neighboring ledgers disagree rather than adding a freestanding cost. The v1.0 rule is stated early because curvature is the easiest term to misunderstand:

Curvature is diagnostic by default. It enters the boundary ledger only when a neighboring-ledger residual, bending elasticity, wall pinning, roughness, or unresolved curvature-stress term is explicitly declared.

A residual curvature term may be introduced as:

$$\mathcal{L}_L^{\text{res}} = \mathcal{L}_L + \lambda_L C_L, \quad (95)$$

only when $\lambda_L C_L$ is explicitly interpreted as bending elasticity, wall pinning, unresolved roughness, curvature stress, or a modeling penalty not already represented by interface, pressure, gravity, wetting, and wall-contact terms.

8 Closure bias and constrained baselines

Closure bias is not a new force. It is the local descent direction of the declared boundary ledger:

$$\mathcal{B}_{\text{closure},L}(s) = -\Pi_{\Omega_L^{\text{adm}}} \nabla_s \mathcal{L}_L(s), \quad (96)$$

where $\Pi_{\Omega_L^{\text{adm}}}$ denotes projection into admissible state directions. This says only: if the system can move in a way that lowers the ledger, that direction is a closure tendency at layer L .

For a simple interface-area proxy:

$$\mathcal{K}_L^\Delta = \gamma [A_L - A_{\min,L}]_+. \quad (97)$$

For a shallow quasi-2D cell with depth h and visible interface length ℓ_L :

$$A_L(t) \approx h \ell_L(t), \quad (98)$$

so:

$$\mathcal{K}_L^\Delta \approx \gamma h [\ell_L(t) - \ell_{\min,L}]_+. \quad (99)$$

If γ is unknown or the material is a mixture, use a proxy:

$$\mathcal{K}_{\text{proxy},L}^\Delta = [\ell_L(t) - \ell_{\min,L}]_+. \quad (100)$$

The baseline is always constrained:

$$A_{\min,L} = \arg \min_{A \in \mathcal{A}_L^{\text{adm}}} \mathcal{L}_L(A). \quad (101)$$

In practice, $A_{\min,L}$ means the lowest accessible layer-readable interface under volume, container, wall, wetting, gravity, time, and measurement constraints. It is not a metaphysical minimum.

As a simple reference baseline, an unconstrained isolated droplet of fixed volume V has a spherical area baseline:

$$R_V = \left(\frac{3V}{4\pi} \right)^{1/3}, \quad A_{\min}^{\text{sphere}} = 4\pi R_V^2, \quad (102)$$

so a simple physical ledger baseline is:

$$\mathcal{K}_{\text{drop}}^{\min} \approx \gamma A_{\min}^{\text{sphere}}. \quad (103)$$

This example is only a baseline reference. Walls, gravity, wetting, mixture composition, temperature, and container geometry change the admissible baseline.

9 Boundary states: distinction, fold-thinning, pop

The core boundary-state ladder is:

$$D_L \rightarrow F_L \rightarrow P_L, \quad (104)$$

where:

$$D_L = \text{sealed distinction / readable boundary}, \quad (105)$$

$$F_L = \text{fold-thinning leak / leaky but unbroken trace gate}, \quad (106)$$

$$P_L = \text{pop / rupture-release event}. \quad (107)$$

A more expanded interaction ladder is:

$$D_L \rightarrow A_L \rightarrow R_L \rightarrow F_L \rightarrow P_L, \quad (108)$$

where A_L is absorption/one-way flux and R_L is rotation/circulation.

A phase indicator for a two-phase boundary may be:

$$\chi(x, t) = \begin{cases} 1, & x \in \Omega_1, \\ 0, & x \in \Omega_2, \end{cases} \quad (109)$$

with layer-readable interface:

$$\partial\Omega_L = \{x : \nabla\chi_L(x, t) \neq 0\}. \quad (110)$$

One-way normal flux is:

$$J_n = J \cdot n. \quad (111)$$

Rotation/circulation can be represented by:

$$\omega = \nabla \times v, \quad (112)$$

$$\Gamma = \oint_C v \cdot d\ell. \quad (113)$$

The pop gate is:

$$\pi_L = H(\Delta B_L - \theta_{B,L}) H(C_{T,L} - \theta_{T,L}), \quad (114)$$

where $H(\cdot)$ is a Heaviside step function, ΔB_L is a boundary-update magnitude, $C_{T,L}$ is a scalar trace-coupling score unless a vector channel is declared, and $\theta_{B,L}, \theta_{T,L}$ are layer-relative thresholds.

State conditions can be stated as:

$$\text{sealed: } C_{T,L} \leq \theta_{T,L}, \quad (115)$$

$$\text{leaky: } C_{T,L} > \theta_{T,L} \quad \text{and} \quad \Delta B_L < \theta_{\text{rupture},L}, \quad (116)$$

$$\text{popped: } \Delta B_L \geq \theta_{\text{rupture},L}. \quad (117)$$

A pop is not every boundary update. It is a boundary update that becomes trace-readable at layer L .

10 Composition-sensitive and reactive ledgers

For mixtures and solutions, material parameters are not fixed constants. They become fields. Let $c_i(x, t)$ be concentration fields and $T(x, t)$ temperature. Then:

$$\rho = \rho(c, T), \quad (118)$$

$$\eta = \eta(c, T), \quad (119)$$

$$\gamma_{12} = \gamma_{12}(c, T, \Gamma), \quad (120)$$

$$\theta = \theta(c, T, \Gamma, G_{wall}). \quad (121)$$

Here Γ denotes interfacial accumulation, such as surfactants, ions, impurities, or reactive species.

A composition-sensitive ledger may be written:

$$\mathcal{L}_L(s, c, T) = \gamma_{12}(c, T, \Gamma) A_{12,L} + \sum_j \int_{\Omega_j} \rho_j(c, T) g z dV + \mathcal{L}_{wall,L}(c, T, \Gamma) + \mathcal{L}_{mix,L}. \quad (122)$$

If reactions are possible, add a reaction/exchange term:

$$\mathcal{L}_{react,L} = \int_{\partial\Omega_L} \sum_i \mu_i(c, T) J_i dA, \quad (123)$$

where J_i is species flux across the boundary. For charged species, electrochemical potential may be represented as:

$$\tilde{\mu}_i = \mu_i + z_i F \phi. \quad (124)$$

This produces a useful theoretical distinction:

$$\text{separating boundary} \subset \text{exchanging boundary} \subset \text{reactive boundary}. \quad (125)$$

Some boundaries merely distinguish. Some permit exchange. Some transform the terms on both sides.

11 Exterior response and black-box boundaries

If a container is opaque or deliberately treated as a black box, CLEF should not pretend to know the internal interface. Instead, the exterior layer reads only returned traces.

Let the hidden internal state be $s_{int}(t)$. Let a probe input be $u_L(t)$, such as a tap. Define the unknown container-fluid transfer operator by:

$$\mathcal{H}_L : u_L(t), s_{int}(t) \mapsto y_L(t). \quad (126)$$

With noise ϵ_L , the exterior response is:

$$y_L(t) = \mathcal{H}_L(u_L(t), s_{int}(t)) + \epsilon_L(t). \quad (127)$$

The tap-response signature can be represented by a transform:

$$R_{tap,L}(\omega) = |\mathcal{F}\{y_L(t)\}|. \quad (128)$$

An exterior state vector may be:

$$B_L^{ext} = \{X_{cm}, P_{cm}, I, R_{tap,L}, D_{slosh}, \omega_{res}\}, \quad (129)$$

where:

$$X_{cm} = \text{center of mass / center of gravity}, \quad (130)$$

$$P_{cm} = M \dot{X}_{cm}, \quad (131)$$

$$I = \text{effective inertia tensor}, \quad (132)$$

$$D_{slosh} = \text{damping/slosh response}, \quad (133)$$

$$\omega_{res} = \text{resonant frequencies}. \quad (134)$$

The theoretical rule is:

A sealed container converts internal boundary geometry into exterior response.

If sight is blocked, touch can still read the boundary by resonance. This is not a completed empirical claim in this paper; it is a theoretical route from hidden boundary state to exterior trace.

12 Layer relevance and stopping rule

CLEF does not require every sublayer detail to be modeled. A sublayer difference matters to a declared layer only when it survives projection into that layer above the noise floor. Let x_k be a sublayer state and $\Pi_{k \rightarrow L}$ be the projection from sublayer k into the declared layer L . A perturbation δx_k matters to L when:

$$d(\mathcal{A}_L(\Pi_{k \rightarrow L}(x_k + \delta x_k)), \mathcal{A}_L(\Pi_{k \rightarrow L}(x_k))) > N_L. \quad (135)$$

If the inequality fails, the sublayer difference may be real but remains background variation for the current claim.

A biological example is mitosis. If the declared question is whether a cell divided into two viable daughter cells, the layer may focus on copied chromosomes, spindle separation, cytokinesis, and daughter-cell viability. Exact enzyme motions and base-level chemistry remain background unless they change chromosome copying, segregation, checkpoint behavior, mutation, or cell viability above the layer's failure threshold. If the question is why a replication fork stalled or why a drug blocked division, the molecular chemistry becomes the declared layer. This example is used only as a scale-relevance illustration; standard accounts of cell-cycle and mitosis mechanisms remain in cell-biology references such as [27].

The rule is:

A detail matters when it changes the answer at the declared layer. Small things stop being small when they survive projection into the question being asked.

13 Worked numerical toy example: heat ledger through a boiler-turbine chain

This toy example is not a physical claim; it demonstrates how a boundary ledger can be carried through a multi-step system using declared terms and proxy values. It is not a reactor design or an engineering model.

This example is schematic. It is not a reactor design and does not validate CLEF experimentally. Its purpose is to show how a CLEF report can declare a layer, observable, baseline, coherence channel, cost/proxy term, and failure condition using simple numbers.

Let a generic heat source feed a boiler-turbine chain. Use arbitrary energy units. The component-scale layer is:

$$L_{heat} = (X, r, N_L, \mathcal{A}_L, W_L), \quad (136)$$

where X is the system energy-flow diagram, r is component-scale resolution, $N_L = 2$ energy units is the ledger noise/rounding floor, $\mathcal{A}_L$ is energy-balance accounting, and W_L is the wall where subcomponent chemistry, blade microstructure, and water molecular details are not included unless they change the component-level balance above N_L .

Suppose:

$$Q_{gen} = 100, \quad Q_{boiler} = 90, \quad (137)$$

$$W_{turbine} = 30, \quad Q_{reject} = 55, \quad \mathcal{K}_L^{\text{loss}} = 5. \quad (138)$$

The component ledger closes because:

$$Q_{boiler} = W_{turbine} + Q_{reject} + \mathcal{K}_L^{\text{loss}} = 30 + 55 + 5 = 90. \quad (139)$$

Define an energy-ledger residual:

$$\epsilon_{E,L} = \left| Q_{boiler} - (W_{turbine} + Q_{reject} + \mathcal{K}_L^{\text{loss}}) \right|. \quad (140)$$

A simple coherence channel is:

$$\Phi_{E,L} = 1 - \text{clip}_{[0,1]} \left(\frac{\epsilon_{E,L}}{N_L} \right). \quad (141)$$

For the values above, $\epsilon_{E,L} = 0$ and therefore:

$$\Phi_{E,L} = 1. \quad (142)$$

A toy CLEF ratio may then be reported as coherence per declared loss-cost unit:

$$\text{CLEF}_{heat,L}^{\text{toy}} = \frac{\Phi_{E,L}}{\mathcal{K}_L^{\text{loss}}} = \frac{1}{5} = 0.2 \text{ toy}^{-1}. \quad (143)$$

This is not an engineering efficiency. It is a toy CLEF ratio showing coherence per declared loss-cost unit in the selected accounting layer. If instead an unaccounted residual of 3 units appeared, then $\epsilon_{E,L} = 3 > N_L$ and the component-level layer would fail its energy-ledger check. A sublayer detail such as tube fouling, heat-transfer degradation, phase instability, or turbine loss would then survive upward because it changes the declared component-scale answer.

A checked toy report could pair the energy ledger with a second channel, such as output stability:

$$\vec{\Phi}_{heat,L} = [\Phi_{E,L}, \Phi_{stable,L}], \quad (144)$$

while the default scalar rule would report the vector first and only collapse it if the two channels are declared independent or appropriately grouped.

14 Possible experimental routes for further refinement

This section is intentionally small. The paper is theoretical. These routes are included to show how CLEF could become accountable to measurement later.

Any CLEF experiment must declare at least one readable boundary, one coherence channel, one cost or proxy-cost term, and one baseline. A compact minimum observable package is:

$$E_{\min,L} = (B_L, \Phi_{j,L}, \mathcal{K}_L^{\Delta,\text{proxy}}, B_L^{\text{base}}), \quad (145)$$

where B_L is the boundary being read, $\Phi_{j,L}$ is the coherence channel, $\mathcal{K}_L^{\Delta,\text{proxy}}$ is the cost or proxy-cost term, and B_L^{base} is the declared baseline. Without all four, the route remains an intuition rather than a CLEF experiment.

14.1 Classroom emulsion route

A middle-school-level demonstration can use water, oil, and a small amount of dish soap or food-safe emulsifier. Oil and water are first observed as separated phases. After a declared perturbation, such as a fixed number of shakes, the mixture is observed at sampled times. The emulsifier lowers the oil/water boundary penalty by sitting at the interface: its water-compatible head faces water, while its oil-compatible tail associates with the oil phase. This lets oil break into smaller droplets that remain suspended longer before coalescing [29, 30].

The minimum layer can be declared as:

$$X = \text{visible oil/water field}, \quad (146)$$

$$r = \text{camera scale or visual aperture}, \quad (147)$$

$$N_L = \text{minimum visible droplet size}, \quad (148)$$

$$\mathcal{A}_L = \text{manual count, grid estimate, or simple image segmentation}, \quad (149)$$

$$W_L = \text{point where droplets are too small or cloudy to read separately.} \quad (150)$$

For a dynamic classroom report:

$$T_L = \{t_0, t_1, t_2, t_3\}, \quad \Pi_L = \text{fixed shake count}, \quad \mathcal{M}_L = \text{visual/camera path.} \quad (151)$$

Possible observables include:

$$Q_{\text{drop},L}(t) = \text{number of readable droplets,} \quad (152)$$

$$Q_{\text{sep},L}(t) = \text{height of separated oil layer,} \quad (153)$$

$$Q_{\text{edge},L}(t) = \text{visible boundary/edge support.} \quad (154)$$

A checked classroom score might use:

$$\vec{\Phi}_L = [\Phi_{\text{seg},L}, \Phi_{\text{sep},L}], \quad (155)$$

where segmentation checks whether droplets are readable and separation height checks whether the emulsion is staying mixed or collapsing back into layers.

Failure conditions should be declared before reporting: lighting changes too much; shaking is not repeatable; droplets fall below the visual noise floor; the mixture becomes too cloudy to segment; or the claim exceeds proxy units. In CLEF language:

Soap does not make oil become water. It changes the boundary so oil can stay divided into many small readable droplets inside water.

14.2 Calibrated two-phase cell

A shallow transparent cell containing two immiscible liquids could approximate:

$$A_{12,L}(t) \approx h\ell_L(t), \quad (156)$$

where h is cell depth and $\ell_L(t)$ is visible interface length. If interfacial tension γ_{12} is known, an approximate creation cost is:

$$\mathcal{K}_{12,L}^{\text{create}} = \gamma_{12}h[\ell_L(t_1) - \ell_L(t_0)]_+. \quad (157)$$

If the material is not calibrated, report proxy values instead:

$$\mathcal{K}_{\text{proxy},L}^{\text{create}} = [\ell_L(t_1) - \ell_L(t_0)]_+. \quad (158)$$

14.3 Lava-lamp or thermally driven boundary proxy

A lava-lamp-like video can expose blob formation, necking, splitting, merging, circulation, and relaxation. Because it is thermally driven and material constants are usually unknown, it should be treated as a proxy domain:

$$\mathcal{K}_{\text{proxy},L}^{\Delta} = [\ell_L(t) - \ell_{\min,L}]_+. \quad (159)$$

It is not a clean physical-value experiment unless geometry, temperature, materials, scale, and interfacial properties are independently calibrated.

14.4 Sealed container tap-response

For an opaque bottle, the hidden liquid level may be inferred through vibration or acoustic response. A declared tap input $u_L(t)$ and returned signal $y_L(t)$ produce:

$$R_{\text{tap},L}(\omega) = |\mathcal{F}\{y_L(t)\}|. \quad (160)$$

This route tests the claim that an internal boundary can become exterior-readable through resonance even when it is not visually accessible.

14.5 Empirical-status note

CLEF is presented here as a theoretical and formal boundary language with proposed empirical anchors. The examples above are not reported as completed experiments in this paper. They are included to show how CLEF can become accountable to measurement by declaring a layer L , observable boundary quantities, proxy costs, baselines, and failure conditions. No physical-value claim is made without calibrated materials, geometry, and measurement data.

15 Conclusion

CLEF begins with the observation that a boundary does not become scientifically accountable merely because it is named. It becomes accountable when a layer, observable, cost, baseline, and failure condition are declared. The core score is simple:

$$\text{CLEF}_L = \frac{\Delta\Phi_L}{\mathcal{K}_L}. \quad (161)$$

The resolution in v1.0 is that this score is paired with checked coherence rules and a mandatory excess state:

$$\mathcal{K}_L^\Delta = \left[\mathcal{K}_L^\sigma - \mathcal{K}_L^{\min} \right]_+. \quad (162)$$

Creation explains how readability was gained. Excess explains what remains loaded in the boundary after it becomes readable.

The boundary ledger replaces vague free-energy language with declared terms. Curvature becomes diagnostic unless a residual role is explicitly declared. Experiments move to the end as future routes, not as the backbone of the paper. The theoretical core is therefore sharper:

Metaphysics can name the boundary. Science has to measure what the boundary does. CLEF is the layer language between those two acts.

A final scale-relevance rule follows from the same structure:

Small things stop being small when they survive projection into the question being asked.

A Anchored readings of spheres as reference bridge

Sphere-anchor language is treated as a reference bridge to the separate spherical-closure branch, not as a new topological claim inside CLEF. The relevant prior paper frames spherical closure through the hairy-ball theorem and Poincare-Hopf interpretation, treating defects, seams, poles, or remainders as layer-permission costs when orientation-bearing patterns attempt global closure on S^2 [28]. CLEF imports only the guarded reading:

$$\text{closed form} + \text{declared reading} \rightarrow \text{selected anchor, seam, pole, or remainder}. \quad (163)$$

The bridge statement is:

A sphere may close geometrically while a reading of the sphere still requires an anchor.

This is analogy and boundary language, not a claim to solve black-hole information problems or no-hair theorem questions.

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AI Tool Assistance Note

This paper was developed through iterative dialogue with AI language-model tools. The author supplied the originating conceptual images and framework direction, including geometric fluid, measurement as a system, boundary readability, spherical closure/anchor intuition, and friction-like interaction as boundary-to-boundary tension. AI tools were used as drafting, mathematical-structuring, terminology-testing, and literature-alignment aids; the author selected, revised, accepted, rejected, and directed the resulting formulations. AI assistance is not treated as authorship, invention, or independent theoretical authority.